

WASP-12b

R-2484

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_140321 · created 2026-08-24T00:42:06+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456738.748671 +/- 3.4e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.085 +/- 0.043
Transit depth (Rp/Rs)^2	0.73 +/- 0.73 [%]
Orbital Inclination (inc)	83.53 +/- 2.81
Airmass coefficient 1 (a1)	0.965 +/- 0.018
Airmass coefficient 2 (a2)	0.038 +/- 0.038
Scatter in the residuals of the lightcurve fit is	1.88 %
Best Comparison Star	#2 - [444, 68]
Optimal Aperture	3.32
Optimal Annulus	8.59
Transit Duration (day)	0.0796 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.085 $\pm$ 0.043 vs 0.1178 (deviation 0.76 σ)
Duration fit vs published	0.0796 d vs 0.125 d (deviation 11.35 σ)
Mid-time O-C	1571.7 min
Depth SNR	2.0
Residual scatter	1.88 %

Light curve

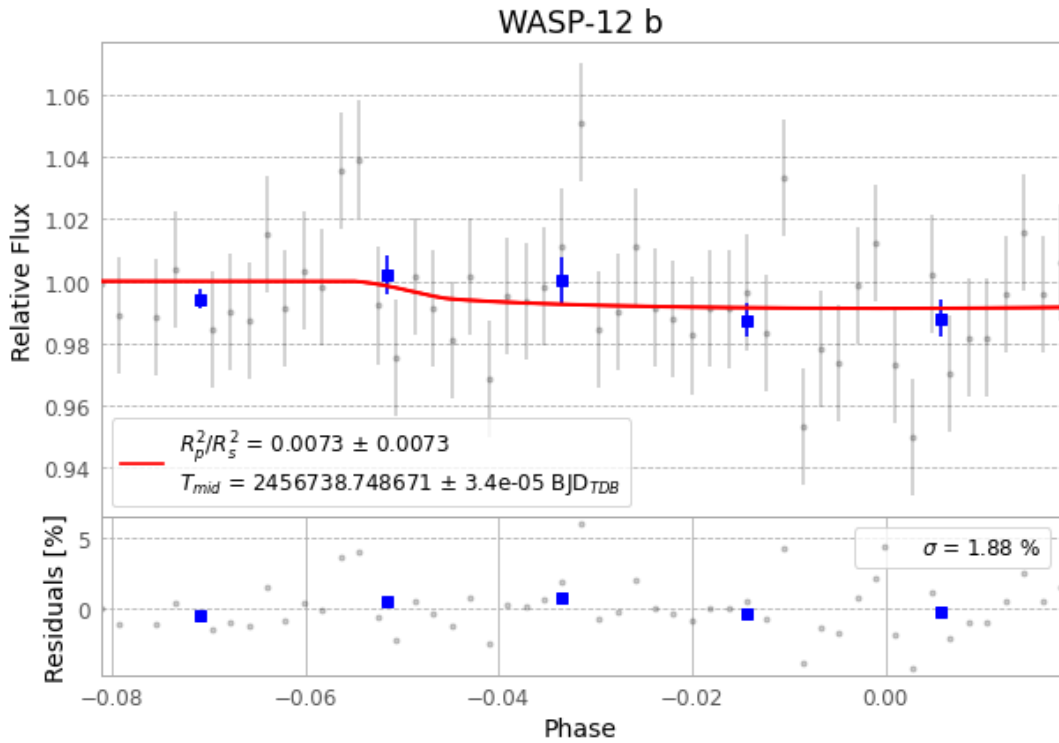


Figure 1 — FinalLightCurve_WASP-12 b_2014-03-21.png (SHA-256 b360c92f9fac9ecb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 293], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 df225ddcd5beda0b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

92 FITS frames (60 MB), WASP-12140322034813.FITS → WASP-12140322082112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-140321.log (a9c241830c9e...), FinalLightCurve_WASP-12 b_2014-03-21.png (b360c92f9fac...), FinalLightCurve_WASP-12 b_2014-03-21.csv (58f47ad99085...)

Compute resources

Wall time 97.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)